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Project Title:	Molecular Genetics of Root Thigmoresponsiveness in Arabidopsis thaliana		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
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Space Biology Special Category:	None		
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Zip Code:	53706-1580	**Congressional District:**	2
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	Masson, Patrick		
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Task Description:	Directional cues within the environment (gravity, touch, light etc) guide plant organs' growth by differential regulation of cell expansion. These tropic movements are coupled to endogenous oscillating movements (nutations) whose origin remains poorly understood. Together, these movements allow plant organs to position themselves for optimal performance. To better understand the mechanisms that govern integrated growth responses to multiple stimuli, we have sought and characterized Arabidopsis thaliana mutations that affect root waving and skewing on tilted hard agar surfaces. Such movements result from a combination of gravity, touch, hydrotropic and other surface-derived stimuli, as well as circumnutations. One of these mutations, named wvd2-1, was shown to alter spiral root growth on agar surfaces and anisotropic cell expansion. The WVD2 protein appears to modulate the arrangement of cortical microtubules in expanding cells. This proposal is aimed at understanding how WVD2 and other related proteins modulate anisotropic cell expansion and root growth behavior. This will be accomplished by identifying and characterizing proteins that regulate or modulate WVD2 function, and by characterizing seven other WVD2-related proteins in Arabidopsis. Plants constitute an essential component of Advanced Life Support Systems for long-term space-exploration missions. However, during space flight, their growth will be guided by cues other than gravity, including nutations. Therefore, we need to better understand some of the mechanisms that will contribute to guiding organ growth under these conditions, a primary objective of this proposal. Also, there is limited space available for plant growth in spacecrafts. WVD2 may provide a good tool to design reversible engineering strategies that would confer a shorter stature to a broad range of plants, yet would not alter seed production or plant's ability to perceive gravity at their final destination. Hence, a thorough characterization of WVD2-dependent pathways is of direct relevance to the Space Biology Program. The objective of this proposal is to characterize the molecular mechanisms that allow WVD2 and related WDL proteins to control root growth behavior in response to mechanical stimuli, anisotropic cell expansion and plant morphogenesis. Our long-term goal is to use the information that will derive from this project to engineer shorter and stockier plants than wild type, which can still produce normal levels of seeds, but are better suited for growth in the limited confines of bioregenerative life support systems during long-term space exploration missions, or that better resist to lodging by wind and rain on Earth or other planets. The WVD2 and WDL genes appear to function not only as key regulators of root growth behavior in response to mechanical stimuli, but also as major determinants in the control of anisotropic cell expansion in all organs of the plant. Therefore, a better understanding of these genes is likely to provide important new insights into how mechanical information in the environment controls cell expansion, and how this affects overall plant morphogenesis, a process of fundamental importance for agricultural production and plant adaptation to its environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Root-growth behavior conditions the ability of plants to populate and grow in very heterogenous and diverse environments. Hence, a better understanding of the molecular mechanisms that govern this process is likely to provide the tools needed to engineer plants that are better able to colonize extreme environments on Earth, and on other planets. At the same time, controlled anisotropic cell expansion is an essential component of plant morphogenesis and behavior in response to directional environmental stimuli. Our studies have allowed the identification of an important new dominant regulator of this process. Because over-expression of WVD2 results in shorter plants with thicker organs, it may be possible to use this engineering strategy to generate crop plants that are more resistant to lodging by wind and rain, as well as plants that are better suited for utilization in spacecrafts, where limited amounts of space are available for plant growth.
	Introduction: As discussed in previous reports, wvd2-1 is a dominant mutation that results from over-expression of a wild-type 23 kD protein containing a novel conserved KLEEK domain with similarity to the TPX2 pfam motif found in several plant and animal unconventional microtubule binding proteins. In Arabidopsis, 7 other genes are predicted to encode KLEEK-containing proteins (WDL1 – 7). wvd2-1 seedlings display an inability to develop root waves and opposite root skewing relative to wild type on tilted hard agar surfaces, enhanced spiral organs growth, altered anisotropic cell expansion and hypersensitivity to osmotic shock, mechanostimulation and light. These phenotypes are accompanied by altered cortical microtubule alignment in epidermal cells of the elongation zone and in peripheral cap cells (Yuen et al., 2003), and also occur in plants over-expressing any one of the 7 WDL genes. Together, these data suggest that WVD2 and other WDL genes have at least partially overlapping functions. We also reported last year that E. coli - expressed WVD2 protein binds to in vitro polymerized microtubules and promotes their bundling, suggesting that WVD2 regulates anisotropic cell expansion and root growth behavior by interacting with, and promoting the bundling of, cortical microtubules in expanding cells. These in vitro observations are consistent with aspects of the wvd2-1 phenotype and with an in vivo study of the dynamic instability of cortical microtubules in this WVD2-over-expressing mutant. These results have been summarized in a paper that was recently published in The Plant Journal (Perrin et al., 2007). We also reported last year that aspects of the wvd2-1-like phenotype, along with novel gene annotations, suggested a possible involvement of brassinosteroids, plant hormones that regulate multiple aspects of plant growth, development and morphogenesis, in the regulation of WVD2. As discussed below, we have now initiated studies aimed at investigating this exciting possibility. Could WVD2 and/or some WDL proteins modulate aspects of brassinosteroid signal transduction in plants? As discussed in last year's progress report, wvd2-1 seedlings display enhanced sensitivity to high light. When light intensity is raised above 120µE.m-2.s-1, mutant seedlings become pale, and their epidermal cells bulge out. Interestingly, annotation of most Arabidopsis WDL genes at the TAIR database (http://www.arabidopsis.org) indicates similarity to rice BRI1-KD-interacting proteins (BIP) 116 and 118 (GB:BAD11344.1). WU-BLAST sequence alignments confirmed similarity throughout most of the sequences between rice BIP116 and Arabidopsis WDL1-3 and WVD2 proteins (4e-17<P(N)<9e-31), and between BIP118 and WDL5, WDL6 and WDL4 (1e-41<P(N)<8e-57). Published biochemical data indicate that rice BIP116 and BIP118 interact with, and are phosphorylated by, the kinase domain (KD) of BRI1, a receptor-like protein kinase that functions as a brassinosteroid receptor (Hirabayashi et al., 2004); pdf at http://www.jspcmb.jp/ Furthermore, WDL1 and WDL7 are regulated by brassinolide, and expression of BRI1 and WDL4 appears co-regulated in a set of 54 experiments (www.genevestigator.ethz.ch).

Task Progress:

Taken together, these data strongly suggest that some of the WDL proteins may be regulated by brassinosteroids through direct interaction with the BRI1 receptor, thereby directly connecting cortical MT remodeling to brassinosteroid action. To start addressing this exciting possibility, we tested the ability of purified, E. coli-expressed, MBP-tagged WVD2 protein to interact with purified, E. coli-expressed, GST-tagged, BRI1 kinase domain (BRI1-KD), using an in vitro pull-down approach. MBP-WVD2 was pulled down by GST-BRI1-KD bound to glutathione-sepharose 4B beads (Amersham), whereas MBP alone was not. Furthermore, neither MBP-WVD2 nor MBP alone could be pulled down by GST bound to glutathione-sepharose beads. We conclude that WVD2 and BRI1-KD interact in this assay, and experiments are in progress to determine if this interaction results in the phosphorylation of WVD2 by BRI1-KD, investigate the target phosphorylation site, and examine its relevance in vivo (Gengxiang Jia and Patrick H. Masson, unpublished results). To assure target specificity, we also built a kinase-dead mutant of BRI1-KD, called GST-BRI1-mKD. This mutant carries a Gly1078Lys mutation in the BRI1-KD, and was previously shown to be an effective negative control in both in vitro binding and phosphorylation experiments (Hirabayashi et al., 2004). This BRI1-mKD gene was inserted into the pGEX-4T-1 vector, and tested for expression in E. coli Origami 2 and BL21 cells. Preliminary results indicate that the GST-tagged BRI1-mKD protein is not produced in Origami 2 cells, but it is produced in large quantities in the BL21 cells, where it can be found in both the soluble form and in inclusion bodies. The soluble BRI1-mKD protein is being tested for its inability to bind to MBP-WVD2 in the in vitro pull-down assay described above (Jia and Masson, unpublished data). To further investigate the possible binding of BRI1-KD to other WDL proteins, we also generated MBP-WDL fusions (similar to the MBP-WVD2 construct described above) and inserted them into the pMAL-c2x vector for expression in E. coli. So far, such expression constructs have been generated for WDL1, WDL2, WDL5 and WDL6. Preliminary optimization experiments indicate that these fusion proteins are poorly expressed in E. coli Origami 2 cells. However, they are highly expressed when introduced into E. coli BL21 cells, from which they can be extracted in the soluble form. Hence, we should be able to test the ability of all four proteins to interact with BRI1-KD in vitro, and be phosphorylated by it, as discussed above.

Overall, these preliminary experiments are encouraging, and suggest that WVD2 and, possibly, other WDL proteins, might directly interact with the BRI1 brassinosteroid receptor, thereby constituting a potential direct link between brassinosteroid reception and microtubule remodeling in plant cells, with possibly direct effects on anisotropic cell expansion and growth behavior. Current work is being aimed at carefully testing the existence of such a functional link using a combination of in vivo immunoprecipitation from plant extracts, identification of potential phosphorylation sites for BRI-KD in WVD2 and WDL proteins, and functional analysis of targeted mutagenezed versions of these genes (Jia and Masson, unpublished results).

Expression and protein localization studies of WDL1-WDL7:

Last year, we reported having generated transgenic plants containing constructs that fuse the WDL genes to GFP, in frame, under their native promoters, or under the strong CaMV 35S promoter (for over-expression), along with constructs fusing the native WDL promoters to GUS. These constructs were aimed at testing expression patterns, protein localization and the phenotypic effects of WDL over-expression in Arabidopsis. We also reported that over-expression of WDL-GFP fusions resulted in phenotypes that are similar to those derived from over-expression of the wild-type WDL proteins, suggesting the GFP moiety in these constructs does not affect protein function.

We propagated multiple independent transformants for each construct, and generated stable transgenic lines. As discussed last year, promoter-GUS fusions revealed complementary expression patterns for WDL1p-GUS and WDL4p:GUS, supporting expression specialization for different members of this gene family. This year, we also used GFP fluorescence to investigate the localization of WDL-GFP proteins in over-expressing plants, using confocal microscopy. Results indicate that all WDL proteins analyzed so far associate with the cortical microtubules in expanding cells of roots, hypocotyls and leaves, supporting a role for all WDL proteins in modulating the organization and/or dynamic instability of the cortical microtubules in expanding plant cells. Current work is aimed at investigating WDL-GFP expression and protein localization in plants that express translational fusion constructs under the control of the native WDL promoters (Jia and Masson, unpublished results).

Summary:

We substantially progressed last year in our functional analysis of the WVD2/WDL gene family in Arabidopsis thaliana. By demonstrating an ability for WVD2 to interact with the kinase domain of BRI1, we have uncovered a potential involvement of WVD2 in directly connecting brassinosteroid action at its receptor with the remodeling of the cortical microtubule network in responding cells, thereby opening the interesting possibility of a direct effect of brassinosteroid action on the dynamic instability of the cortical microtubules with potentially important and direct effects on the regulation of anisotropic cell expansion and growth behavior. The possibility that other WDL proteins might mediate similar actions while displaying distinct expression patterns is being investigated using in vitro analysis of potential interactions and phosphorylation, an examination of expression patterns and protein localization, and a phenotypic investigation of over-expression and knockout mutants. This next phase of our investigations should, next year, yield important breakthroughs in our understanding of the molecular mechanisms that govern anisotropic cell expansion and root growth behavior in response to endogenous and environmental cues in plants.

These analyses should lead to substantially better understanding of the partially overlapping functions played by these genes in plant growth, development and response to their mechanical and biotic environment. In the long run, the corresponding information will help designing and engineering shorter and stockier plants for utilization in the limited confines of Bioregenerative Life Support systems during long-term space exploration missions.

Cited literature

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Bibliography Type:	Description: (Last Updated: 07/03/2024)
Articles in Peer-reviewed Journals	Perrin R, Wang Y, Yuen C, Will J, Masson PH. "WVD2 is a novel microtubule-associated protein in Arabidopsis thaliana." Plant J. 2007 Mar;49(6):961-71. PMID: 17319849 , Mar-2007